

Cloud Learning Roadmap (Practical + Conceptual)

Goal: Learn cloud computing in a hands-on, application-oriented way focused on software engineering, deployment, and DevOps.

Recommended YouTube Channels

freeCodeCamp - Complete AWS courses and cloud fundamentals.

Tech With Lucy - Practical AWS projects and architecture explanations.

Be A Better Dev - AWS concepts explained simply.

KodeKloud - Excellent hands-on DevOps and cloud labs.

John Savill's Technical Training - One of the best Azure learning resources.

TechWorld with Nana - Docker, Kubernetes, CI/CD, DevOps, and cloud deployments.

Learning Roadmap

Phase 1: Cloud Fundamentals (1 Week)

Learn cloud computing, IaaS/PaaS/SaaS, regions, availability zones, virtual machines, networking basics, IAM, EC2, and S3. Project: Host a static website on S3.

Phase 2: Networking (1 Week)

Learn VPC, subnets, internet gateways, route tables, security groups, and load balancers. Project: Deploy a React application on EC2.

Phase 3: Databases (1 Week)

Learn RDS, DynamoDB, backups, and replication. Project: Move a MERN database to the cloud.

Phase 4: Containers (2 Weeks)

Learn Docker and Docker Compose. Project: Dockerize your MERN Job Portal.

Phase 5: DevOps (2 Weeks)

Learn GitHub Actions, Jenkins, and CI/CD pipelines. Project: Automatic deployment from GitHub.

Phase 6: Kubernetes (2–3 Weeks)

Learn Pods, Deployments, Services, Ingress, and Scaling. Project: Deploy your MERN application on Kubernetes.

Projects to Build

- Deploy your MERN Job Portal on AWS (EC2, S3, RDS, Nginx)
- Netflix Clone Deployment using React, Docker, and AWS
- URL Shortener using Spring Boot, Redis, and AWS
- CI/CD Pipeline Project using GitHub Actions, Docker, and AWS
- Microservices Deployment using Spring Boot, Docker, and Kubernetes

60-Day Plan

Days 1–15: AWS Basics, IAM, EC2, S3, VPC

Days 16–30: Application deployment, databases, load balancers

Days 31–45: Docker, GitHub Actions, CI/CD

Days 46–60: Kubernetes and end-to-end deployment project